Quiz (Soild Mechanics-BCE23301T)

Time: 20 Minutes
Marks: 30
****All questions having equal marks**
Total Questions:15
Semester: Third
Civil Engineering Department, Girijananda Chowdhury University (GCU), Assam

* Indicates required question

1. Email *

2. **Roll No.** *

3. **Name** *

4. 1. Stress is — 2 points

Mark only one oval.

- External force
- Internal resistive force
- Axial force
- Radial force

5. 2. Tensile Strain is — 2 points

Mark only one oval.

- Increase in length / original length
- Decrease in length / original length
- Change in volume / original volume
- All of the above

6. 3. Compressive Strain is 2 points

Mark only one oval.

- Increase in length / original length
- Decrease in length / original length
- Change in volume / original volume
- All of the above

7. 4. Hooke's law is applicable within 2 points

Mark only one oval.

- Elastic limit
- Plastic limit
- Fracture point
- Ultimate strength

8. 5. If a material expands freely due to heating it will develop 2 points

Mark only one oval.

- Thermal stress
- Tensile Stress
- Compressive Stress
- No Stress

9. 6.Thermal Stress in a composite bar **DOES NOT** depend on which of the following factors? 2 points

Mark only one oval.

- Coefficient of thermal expansion
- Temperature Change
- Area of Crossection
- Modulus of elasticity

10. 7. Young’s Modulus of elasticity is 2 points

Mark only one oval.

- Tensile stress / Tensile strain
- Shear stress / Shear strain
- Tensile stress / Shear strain
- Shear stress / Tensile strain

11. 8. The relationship between Young’s modulus (E), Bulk modulus (K) and Poisson’s ratio (μ) is given by 2 points

Mark only one oval.

- E=2K(1-2μ)
- E=3K(1-2μ)
- E=2K(1-3μ)
- E=5K(1-3μ)

12. 9. The ratio of lateral strain to linear strain is called 2 points

Mark only one oval.

- Modulus of Elasticity
- Modulus of Rigidity
- Bulk Modulus
- Poisson’s Ratio

13. 10. Which of the following is a dimensionless quantity? 2 points

Mark only one oval.

- Shear stress
- Poison’s ratio
- Strain
- Both (b) and (c)

14. 11.In the context of principal planes and principal stresses, the stresses on oblique section are determined by which of the following method(s): 2 points

1. Analytical Method
2.Graphical Method

Mark only one oval.

- Only I
- Only II
- Neither I nor II
- Both I and II

15. 12. Which of the given option is **NOT** correct for Mohr's Circle? 2 points

I) Y-axis represents normal stress
II) X-axis represents shear stress
III) Half of the diameter represents maximum shear stress

Mark only one oval.

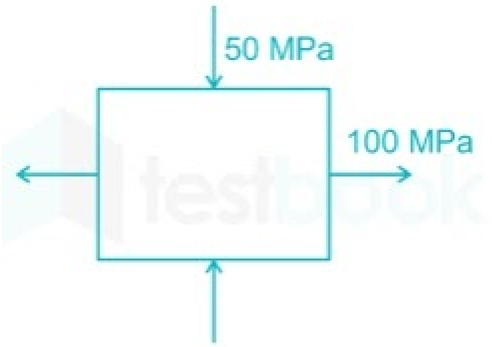
- II & III
- I & III
- I,II & III
- I & II

16. 13. The angle between Major Principal Plane and Minor Principal Plane for a strained body is 2 points

Mark only one oval.

- 45°
- 30°
- 60°
- 90°

17. 14.For the state of stress shown in the below figure,find the principal stresses.Major and Minor stresses are respectively as follows: 2 points



Mark only one oval.

- Major Principal Stress= 150 MPa, Minor Principal Stress= 75 MPa
- Major Principal Stress= 100 MPa, Minor Principal Stress= -50 MPa
- Major Principal Stress= -100 MPa, Minor Principal Stress= 50 MPa
- Major Principal Stress= 150 MPa, Minor Principal Stress= 100 MPa

18. 15.The Planes, which have no shear stress are known as 2 points

Mark only one oval.

- Tangential Planes
- Nominal Planes
- Normal Planes
- Principal Planes